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25 August 1952
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27 November 1950

MILITARY STANDARD

NONDESTRUCTIVE TESTING SYMBOLS

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MUNITIONS BOARD STANDARDS AGENCY
DEPARTMENT OF DEFENSE
Washington, D. C.

25 August 1952

Nondestructive Testing Symbols
MIL-STD-23A

1. This standard has been approved by the Departments of the Army, the Navy, and the Air Force, and establishes the symbols for use on drawings to specify nondestructive tests for determining the soundness of materials.

2. The Munitions Board Standards Agency on 25 August 1952 approved the revision of this standard for printing.

3. In accordance with established procedure the Munitions Board Standards Agency has designated the Ordnance Corps, Bureau of Ships, and the Department of the Air Force, respectively, as Army-Navy-Air Force custodians of this standard.

4. This revised standard is mandatory for use effective 25 August 1952. Where repeated deviation from the approved Military Standard is granted by a department, a report shall be made to the Munitions Board Standards Agency by that department with the reasons for deviation.

5. Recommended corrections, additions, or deletions should be addressed to the Director, Munitions Board Standards Agency, Washington 25, D. C.

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I Basic Nondestructive Testing Symbols

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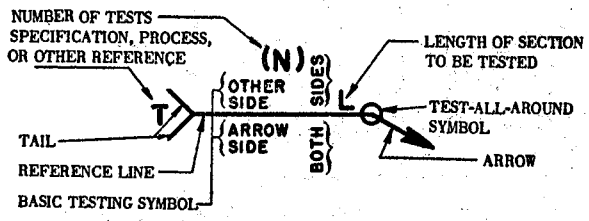
1. BASIC SYMBOLS

1.1 BASIC TESTING SYMBOLS. Basic Nondestructive Testing symbols shall be as shown in table I.

TYPE OF TEST	SYMBOL
RADIOGRAPHIC	RT
MAGNETIC-PARTICLE	MT
PENETRANTS	PT
ULTRASONIC	UT

1.2 ELEMENTS OF THE TESTING SYMBOL. The assembled testing symbol consists of the following elements. Such of these elements as are necessary shall be used and shall have standard locations with respect to each other as shown on figure 1.

Reference Line
Arrow
Basic Testing symbol
Test-all-around symbol
(N) Number of Tests
Tail
Extent of test
Specification, process, or other reference

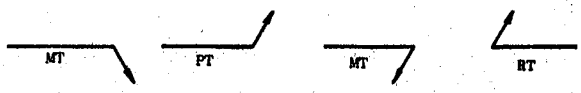


2. GENERAL PROVISIONS

2.1 SIGNIFICANCE OF ARROW. The arrow shall connect the reference line to the part to be tested. The side of the part to be tested, to which the arrow points shall be considered the arrow side of the part. The side opposite the arrow side of the part shall be considered the other side.

2.2 LOCATION OF TESTING SYMBOL.

2.2.1 Tests to be made on the arrow side of the part shall be indicated by the test symbol on the side of the reference line toward the reader as follows:

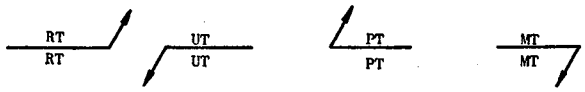


2.2.2 Tests to be made on the other side of the part shall be indicated by the test symbol on the side of the reference line away from the reader as follows:

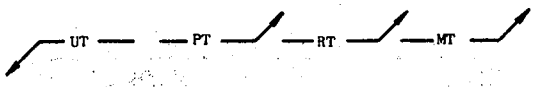


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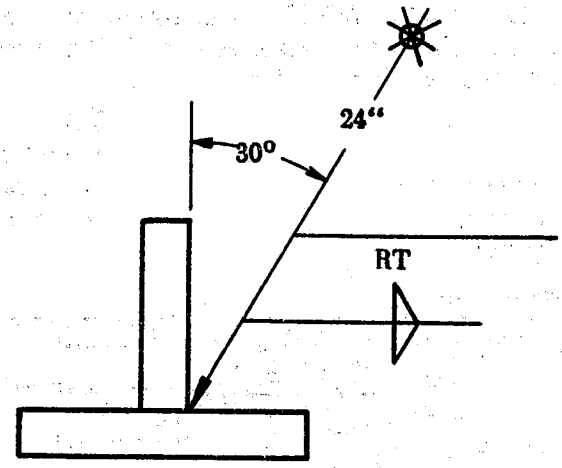
2.2.3 Tests to be made on both sides of the part shall be indicated by test symbols on both sides of the reference line, as follows:



2.2.4 When nondestructive testing symbols have no arrow or other-side significance, the testing symbols shall be centered on the reference line as follows:

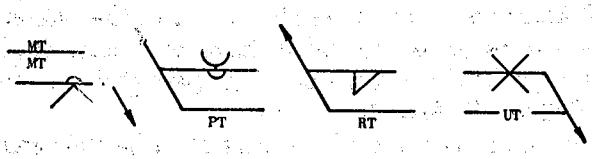


2.3 **DIRECTION OF RADIATION.** When specified, the location of the source of radiation and the direction of radiation shall be shown in conjunction with radiographic testing symbol. The location of the source of radiation shall be indicated by a symbol located on the drawing at the desired source of radiation, connected, and oriented as necessary by dimensions, as shown on figure 2.

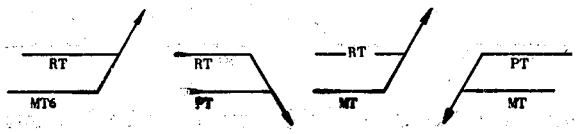


2.4 **COMBINATION OF NONDESTRUCTIVE TESTING SYMBOLS AND WELDING SYMBOLS**

2.4.1 Nondestructive testing symbols and welding symbols may be combined as follows:



2.4.2 Nondestructive testing symbols may be combined as follows:



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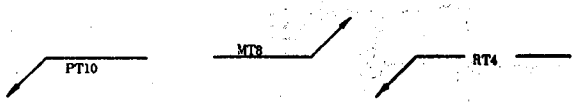
2.5 USE OF REFERENCES. Specifications, process, classification, or other references need not be used on testing symbols when the testing procedure is prescribed elsewhere. When a specification, process, classification, or other reference is used with a testing symbol, the reference shall be placed in the tail, as follows:



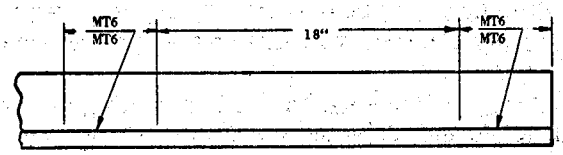
3. METHODS OF SPECIFYING EXTENT OF NONDESTRUCTIVE TESTS

3.1 SPECIFYING LENGTH OF SECTION TO BE TESTED

3.1.1 To specify tests of welds or parts where only length of section need be considered, the length in inches shall be shown to the right of the basic test symbol, as follows:

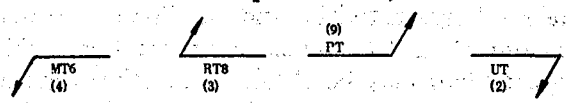


3.1.2 To show the exact location of a section to be tested as well as its length, dimension lines shall be used as follows:

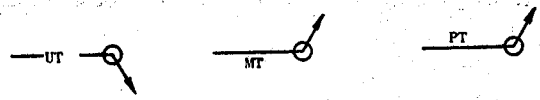


3.1.3 When the full length of a part is to be tested, no length dimension need be shown on the testing symbol.

3.2 SPECIFYING NUMBER OF TESTS. To specify a number of tests to be taken at random, the number of desired tests shall be shown in parenthesis, as follows:

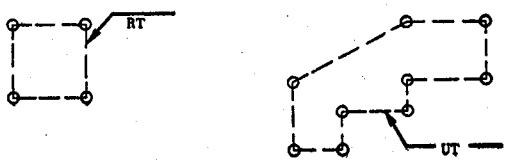


3.3 SPECIFYING TESTS MADE ALL AROUND A PART. To specify tests to be made all around a part, the test-all-around symbol shall be used, as follows:



3.4 SPECIFYING TESTS OF AREAS. When required, nondestructive tests of areas shall be indicated by one of the following methods:

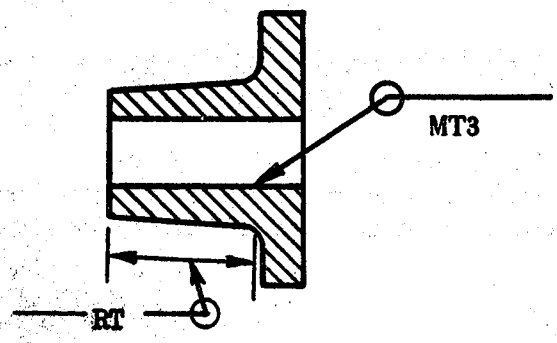
3.4.1 For nondestructive testing of an area represented as a plane on the drawing, the area to be tested shall be enclosed by straight broken lines having a circle at each change of direction. The testing symbol specifying the kind of nondestructive test shall be used in connection with these lines, as shown below. When necessary, these inclosures shall be located by dimensions.



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3.4.2 For nondestructive testing areas of revolution, the area shall be indicated by using the test-all-around symbol as follows:

The symbol at left indicates that the bore of the flange is to be subjected to a magnetic-particle examination for a distance of three inches from the face, all the way around.



The above symbol indicates an area of revolution to be subjected to radiographic examination where dimensions are not available on drawing.

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For the Marine Corps, to the Quartermaster General, Headquarters, U. S. Marine Corps, Washington 25, D. C.

For Department of the Air Force activities and Air Force contractors, to the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Ohio.

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